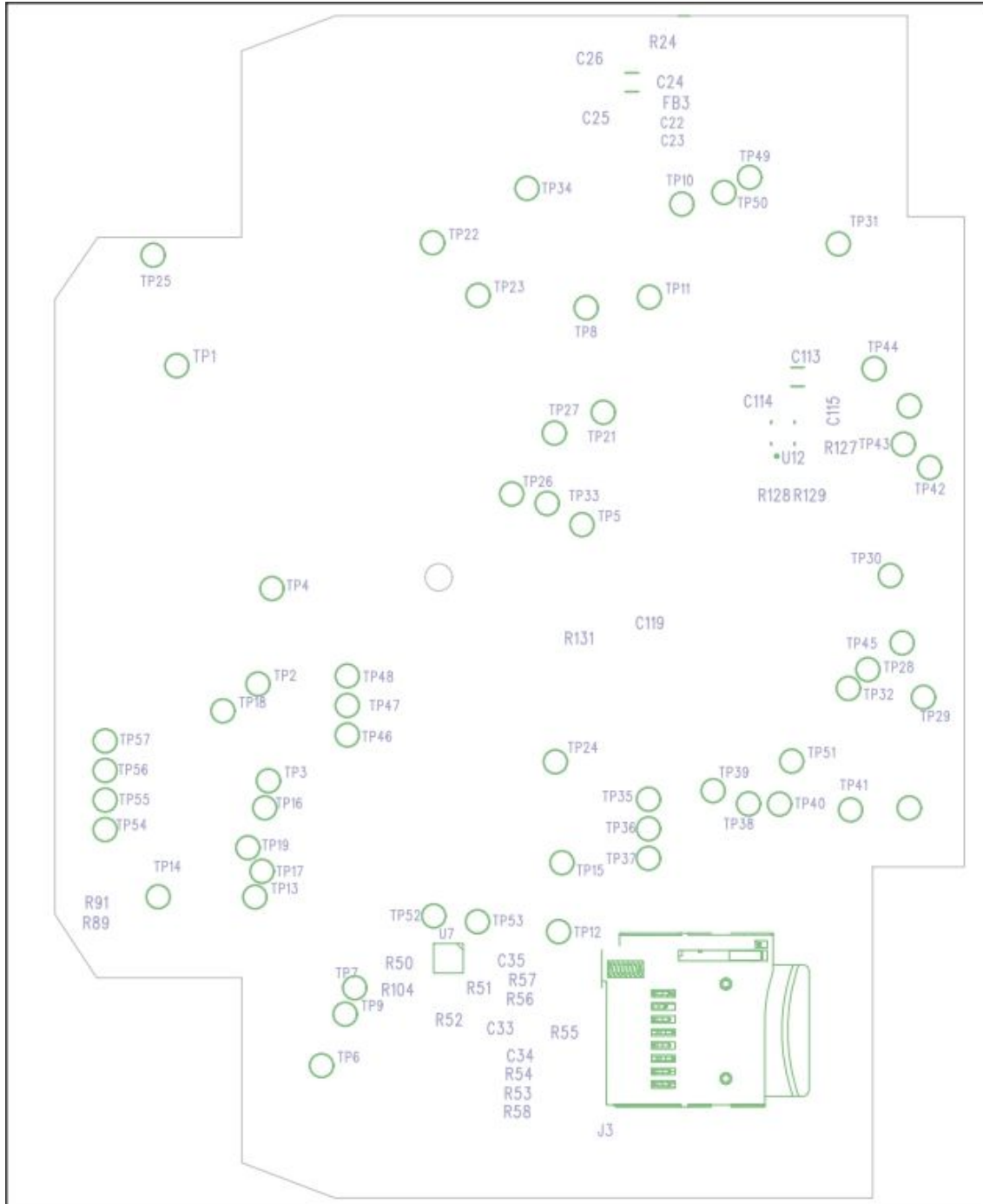


Kello Firmware programming guide

1. Main PCBA description

The main pcba embeds the STM32 host MCU, the Libre WiFi module, the Bluetooth module, the audio DSP and audio PA.

Several test points are available on the bottom side of the pcb.



Main PCBA v7a

interface	test points	parameters
UART STM32 MCU	TP35,TP36,TP37	115200/8/N/1
UART LIBRE WIFI MODULE	TP46,TP47,TP48	57600/8/N/1

SWD STM32 MCU	TP38,TP39,TP40,TP41	
USB LIBRE WIFI MODULE	TP54,TP55,TP56,TP57	

2. Firmware programming

WiFi module image (USB)

An factory image “Isimage” is to be copied to the root of a USB flash drive and a command is sent on the WiFi module UART.

Hardware	Software
USB flash drive FAT32 formatted	“Isimage” factory image copied to the root of the USB flash drive. The factory image “Isimage” contains: • Libre u-boot • Libre BSL • Libre env variables (NV-items) • Libre Linux Application

Interface	Command	Example
UART LIBRE	fwupgrade	<pre> root@libre:/ # fwupgrade ... FENV Flashed!!!! APP image found flashing from /mnt/Application Libre APP Image0000000035303633 Writing from /mnt/Application to Application ... Progress:= 1%write offset 0x680, value 0x10 ... Progress:=100%write offset 0x680, value 0x10 write offset 0x6a8, value 0x1 ... Application Flashed!!!! HOST image found flashing from /mnt/Host Host Image Datasize to uart packet =5 write offset 0x680, value 0x10 ... Host Image ready: Size=71752 Setting packet size to 128 write offset 0x680, value 0x10 ... Packet number=1 </pre>

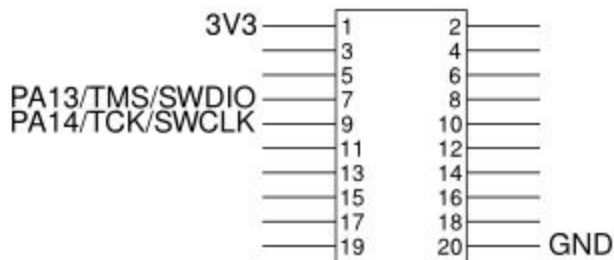
		<pre> ... Packet number=50 xmodem_status=71808 xmodem success:71808 write offset 0x680, value 0x10 ... Sending Reboot request Waiting for Reboot command from host read_from_UART heRestarting system. </pre>
UART LIBRE	getenv Firmware_version	<pre> root@libre:/ # getenv Firmware_version ENV: Value found Firmware_version : LS1.5 </pre>
UART LIBRE	getenv Hardware_version	<pre> root@libre:/ # getenv Hardware_version ENV: Value found Hardware_version : LS1.5 </pre>
UART LIBRE	getenv fwdownload_xml	<pre> root@libre:/ # getenv fwdownload_xml ENV: Value found fwdownload_xml : http://dev.sunup.co/lwfw/ota/kello:1/D0AA06/b </pre>
UART LIBRE	netcfg	<pre> root@libre:/ # netcfg p2pctrl0 UP 0.0.0.0/0 0x00001003 d8:f7:10:d0:aa:06 lo UP 127.0.0.1/8 0x00000049 00:00:00:00:00:00 eth0 DOWN 0.0.0.0/0 0x00001002 00:0c:43:e1:76:29 wlan0 UP 0.0.0.0/0 0x00001003 d8:f7:10:d0:aa:06 </pre>
UART STM32	hello	<pre> product_id: kello.1 device_serial: Kello_D0AC44 mac: D8F710D0AC44 btmac: 7FD6C9 firmware_version: D0AC44:20170306_182311_a5d5dea libre_firmware_version: p5063 bt_firmware_version: 150 mcu_loader_version: 9 ASCII: \n0c00\r tunnel_cmd.sh hello -> "%s:%s:%s:%s:%s:%s" % (device_friendly_name, product_id, device_serial, firmware_version, libre_firmware_version, mcu_loader_version) 000:Kello_D0AC44:kello.1:D0AC44:20170306_182311_a5d5 dea:p5063:9 </pre>

MCU firmware (SWD)

The MCU firmware can be bundled into the WiFi module image for OTA upgrade but this is not the case for the factory image. The reason is that the MCU bootloader can only be programmed through SWD and the MCU firmware is programmed at the same time.

Hardware	Software
ST-LINK/V2 programmer	STM32 ST-LINK utility (CLI tool) openocd 0.9.0

ST-LINK/V2 20pin connector



We're using a STLink v2 to program the STM32F070CB.

We use opensource "openocd" 0.9.0 on linux to program it via the SWD port. It should be possible to use it on windows.

```
# program application + its signature for loader (~5s)
openocd -f interface/stlink-v2.cfg -f target/stm32f0x.cfg -c 'init; program
factory-firmware-*/kello.1-app+sign-*.hex verify; exit'1
```

```
# program secure loader + lock flash for read by SWD (~1s) we could add a serial
number/private key in this step
openocd -f interface/stlink-v2.cfg -f target/stm32f0x.cfg -c 'init; halt; program
factory-firmware-*/kello.1-loader-*.hex verify; stm32f1x lock 0; shutdown'
```

```
# In case you need to set the MCU in its initial erased state :
openocd -f interface/stlink-v2.cfg -f target/stm32f0x.cfg -c "init ; halt ; stm32f1x unlock 0;
stm32f1x mass_erase 0; exit;"
```

MCU serial trace :

```
kello_loader v_a 1
libre_in_app_mode: 0x0
I/O RMID CT CMD ST CRC LEN DATA
```

Bluetooth LSBT

Is shipped with a final image already. It will be 1.06 or 1.10, still to be determined.